SCFDet: Selective Cross-Modal Fusion Network for RGB-Infrared Object Detection in Remote Sensing

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***Abstract*—RGB-infrared object detection in remote sensing is expected to enable all-weather robust detection by combining visible light texture information with infrared light's thermal radiation properties. However, significant distributional and semantic disparities between the two modalities make effective complementary information integration difficult. Moreover, existing methods primarily emphasize cross-modal feature alignment or interaction and largely neglect the enhancement of the fused representations themselves. This oversight leads to underutilization of complementary information in fused features, ultimately hindering further improvement in model performance. To address these issues, we propose the selective cross-modal fusion network (SCFNet), a novel framework for RGB-infrared object detection that leverages a selective cross-modal fusion strategy. First, we design the selective mutual-guided feature alignment (SMA) module, which employs a bidirectional cross-attention mechanism for precise feature alignment, combined with a Top-K strategy to focus on salient regions. Second, we introduce the enhanced expert feature representation (EER) module, which utilizes a sparse Mixture-of-Experts (MoEs) architecture to dynamically select optimal feature transformation paths, coupled with spatial attention modulation to enhance robustness. Third, we propose the cross-channel aware feature enhancement (CAE) module, which models channel-wise dependencies via consistency evaluation, amplifying semantically coherent features and suppressing conflicts. Extensive experiments on the VEDAI and DroneVehicle datasets demonstrate that our method achieves state-of-the-art performance with 80.45% and 85.67% mAP_{50}, respectively.**

***Index Terms*—Object detection, remote sensing, RGB-infrared.**

I. Introduction

ADVANCES in airborne and spaceborne sensors, along with generative techniques [1], [2], [3], [4], have significantly reduced the difficulty of acquiring high-spatial-resolution remote sensing images. This has provided unprecedented data support for fields such as urban fine-grained management, precision agricultural monitoring, and rapid disaster assessment. Consequently, the application of remote sensing technology is transitioning from traditional large-scale land use classification and change monitoring to deeper stages of interpretation, including individual target recognition, state analysis, and behavioral understanding of specific ground objects.

Among these advanced tasks, RGB-infrared object detection has emerged as a particularly promising approach to enhance perceptual robustness in complex environments. However, despite its great potential, this technology still faces a series of severe challenges. First, visible light and infrared images originate from different physical imaging mechanisms, with significant differences in feature distribution, resolution, and information structure [5]. Effective fusion strategies are required to enable collaboration between the two modalities and achieve information complementarity. Second, the dual-modal data contain a substantial amount of redundant information [6], and the difficulty in adaptively filtering and weighting this information poses a core challenge to improving model performance.

Researchers have developed a variety of deep learning-based frameworks for RGB-infrared object detection, which can be categorized as either two-stage [7], [8], [9] or single-stage [10], [11], [12]. These frameworks explore fusion strategies at different levels, including input-level [13], feature-level [14], and decision-level [15]. Among these, feature-level fusion enables cross-modal feature interaction in the deep layers of the network. This approach not only retains richer semantic information compared to input-level fusion methods but also allows for earlier and more comprehensive information complementarity than decision-level fusion methods. The exceptional flexibility of feature-level fusion has made it highly favored in multimodal object detection.

Although numerous methods based on feature-level fusion [16], [17], [18] have been proposed and have advanced the field, current research predominantly focuses on innovations in fusion structures while lacking enhancement and optimization of the fused features themselves. This limitation hinders the model's ability to fully exploit and utilize cross-modal information, thereby restricting further improvement in detection performance.

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Driven by the requirements of RGB-Infrared object detection and the limitations of existing feature-level fusion methods, we propose a selective cross-modal fusion network (SCFNet), an architecture designed based on the principle of selective cross-modal fusion, which emphasizes not only cross-modal interaction but also the enhancement and refinement of fused features. We validate our experiments on two typical datasets: 1) VEDAI [19] and 2) DroneVehicle [8], and compare our model with state-of-the-art RGB-Infrared object detection models to demonstrate its effectiveness. In summary, the main contributions of this work are as follows.

1) We propose the selective mutual-guided feature alignment (SMA) module, which employs a bidirectional cross-attention mechanism to achieve precise spatial and semantic alignment between two modalities. It further incorporates a salient region selection mechanism based on a Top-K strategy to promote efficient fusion by focusing on the most discriminative feature regions.
2) To adaptively enhance feature representation against varying modality characteristics, we introduce the enhanced expert feature representation (EER) module. It utilizes the sparse Mixture-of-Experts (MoEs) to dynamically select optimal feature transformation paths and further incorporates spatial attention modulation to facilitate effective cross-modal feature interaction, boosting adaptability in diverse environments.
3) Aiming for refined feature fusion through inter-channel relationships, we design the cross-channel aware feature enhancement (CAE) module. It explicitly models channel-wise dependencies through consistency evaluation, which amplifies semantically coherent information and suppresses feature conflicts across channels, leading to more discriminative and robust fused representations for cross-modal tasks.

II. Related Work

A. Single-Modal Object Detection in Remote Sensing

The continuous advancement of high-resolution imagery has provided an unprecedented data foundation for remote sensing object detection tasks. Among these, single-modal object detection methods based on specific image modalities have matured significantly, proving their effectiveness in various specialized applications.

These single-modal approaches are often tailored to the characteristics of a specific data type. Li et al. [20] proposed the large selective kernel network (LSKNet), which employs a spatial selection mechanism to adaptively adjust the receptive field, while Cai et al. [21] introduced the poly kernel inception network (PKINet) that leverages multiscale convolutional kernels and a context anchor attention mechanism. Both approaches are designed to address the significant scale variations of targets in optical remote sensing detection. Wang et al. [22] proposed FDI-YOLO, a model with a reversible structure and cross-scale feature interaction, to enhance feature extraction in SAR images plagued by speckle noise and complex backgrounds, thereby boosting detection accuracy and robustness. Besides, in infrared object detection, Liu et al. [23] addressed the issue of low target saliency by proposing a scale- and location-sensitive (SLS) loss and a multiscale head network (MSHNet) to boost discrimination capability. Further advancing feature utilization, Li et al. [24] designed a dense nested attention network (DNA-Net) with a dense nested interactive module and a channel-spatial attention mechanism to maintain feature integrity across network layers.

The above methods highlight the capability of single-modal detection in leveraging the advantages of its specific imaging source. However, its fundamental limitations are equally rooted in this characteristic. In scenarios with insufficient lighting, harsh weather, or when the target has a similar texture compared to the background, the model's performance can drop significantly. This bottleneck has given rise to an urgent need for multimodal techniques, particularly RGB-infrared fusion technology.

B. RGB-Infrared Object Detection in Remote Sensing

RGB-infrared object detection aims to overcome the limitations inherent to single-modality sensing. Its core principle is to leverage the complementary information from both infrared and visible light images to achieve stable and robust all-weather detection performance. In modern multimodal object detection, fusion methods can be broadly categorized into three levels: 1) input; 2) feature; and 3) decision. Feature-level fusion is widely adopted for its superior performance. This paradigm operates by enabling cross-modal features to interact within the network's intermediate layers. By embedding fusion modules into the network architecture, the model can learn more discriminative joint feature representations, thereby effectively improving detection performance.

Researchers have made many improvements based on feature-level fusion. For example, Liu et al. [25] designed a prior-enhanced MoEs fusion network, which leverages a dynamic gating mechanism guided by image entropy and grayscale histograms to adaptively fuse multimodal features under varying environmental conditions. To enhance the network's representational capacity, several studies have adopted specialized attention-based fusion architectures. Liu et al. [26] developed a multibranch progressive fusion network that effectively extracts and integrates complementary information through a global-local synergistic attention module and a complementary information filter. Similarly, Wang et al. [27] proposed YOLOFIV, which leverages a dual-stream backbone with an attention mechanism to hierarchically fuse features through concatenation in shallow-middle layers for preserving modality-specific details and additive fusion in deeper layers for enhancing shared characteristics, all while dynamically attending to critical features. Meanwhile, Shen et al. [28] introduced ICAFusion featuring an iterative cross-attention mechanism, which enables efficient global complementary information extraction through bidirectional enhancement modules and a parameter-sharing strategy without computational overhead. In addition to attention paradigms, researchers have introduced graph neural networks as an alternative approach. Zhang et al. [29] proposed GAANet, which models multimodal features as graph nodes through a graph interaction fusion module. This architecture

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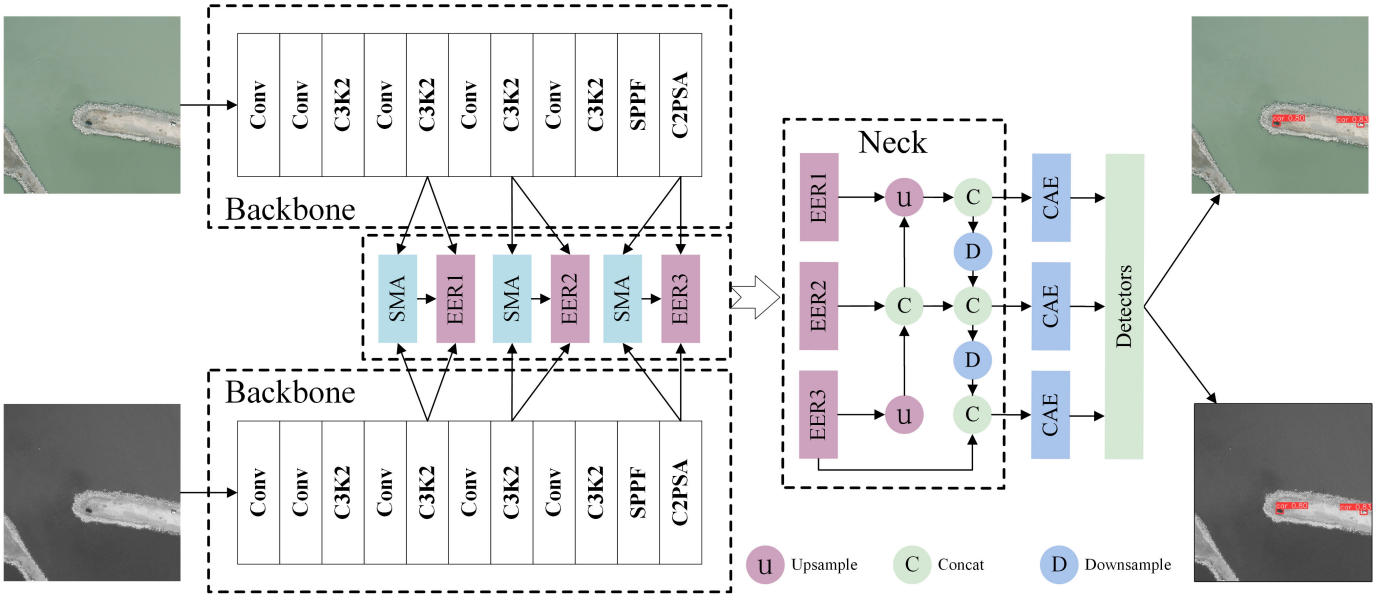


Fig. 1. Overall framework of SCFDet.

achieves global perception and local calibration via information propagation and alignment mechanisms, significantly boosting detection performance on weakly aligned datasets while substantially reducing model parameters. Beyond conventional attention-based and graph-based fusion paradigms, recent studies have begun to explore state-space models like Mamba [30] for multispectral detection. Zhou et al. [7] proposed DMM, a disparity-guided Mamba framework that addresses intermodal and intramodal discrepancies through cross-modal fusion, target-aware attention, and auxiliary supervision. Zhu et al. [31] presented WaveMamba, which integrates discrete wavelet transform with Mamba for RGB-infrared fusion. It separately processes low-frequency features via a Mamba-based fusion block with channel swapping and gated attention, while enhancing high-frequency features through an absolute maximum fusion strategy.

However, existing research predominantly focuses on designing sophisticated fusion architectures while overlooking the optimization of the quality of the fused features. The latent value within the cross-modal data remains underexploited, limiting further performance improvements.

C. MoEs Network

MoEs is a model architecture designed to scale up model capacity efficiently without proportionally increasing computational cost. This method is built upon a collection of expert subnetworks, complemented by a gating network that dynamically assigns each input to the most suitable experts. This paradigm allows different experts to handle different types of data, thereby enhancing the model's representational capability.

Recently, MoE has not only seen significant success in large language models [32], [33] but has also been increasingly applied in the field of computer vision. Specific to vision tasks, recent studies have successfully applied MoE to image classification [34], video understanding [35], and multimodal representation learning [36]. In the field of multimodal object detection, MoE presents a promising direction for managing and fusing data from different modalities. Recent work has begun to explore its usage for feature enhancement and adaptive fusion in RGB-thermal object detection [37] and RGB-D scene understanding [38]. By leveraging specialized experts, MoE facilitates targeted processing of distinct modalities, thereby improving cross-modal alignment and robustness in complex environments.

III. Methodology

A. Overall Architecture

The overall architecture of SCFNet, built upon the YOLO11 framework, is illustrated in Fig. 1. It consists of a dual-stream backbone for extracting RGB and infrared features, a neck for aggregating and transforming features across scales, and detection heads for final object prediction. To achieve selective cross-modal fusion, we sequentially introduce three core modules: the SMA module employs a bidirectional cross-attention mechanism to achieve feature alignment while focusing on critical regions; the EER module utilizes the sparse MoEs network to dynamically select optimal feature transformation paths and enhance representation robustness; and the CAE module optimizes inter-channel feature interactions through consistency evaluation. The implementations of the SMA, EER, and CAE modules are elaborated in Sections III-B–III-D, respectively.

B. Selective Mutual-Guided Feature Alignment

In multimodal object detection, the features of the two modalities not only have distribution differences and semantic misalignment issues, but more importantly, not all regional features contribute equally to the detection task. Directly performing global feature fusion can lead to information redundancy or even noise interference.

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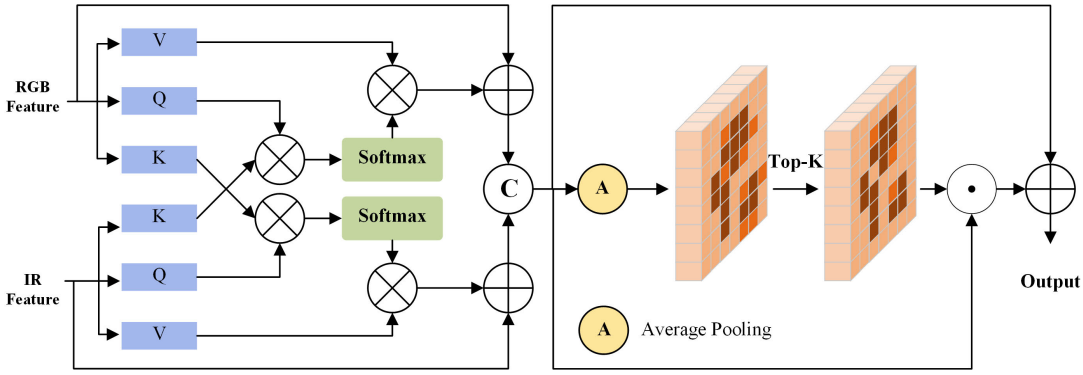


Fig. 2. Architecture of SMA. ⊗ denotes the matrix multiplication, ⊕ represents the element-wise addition, and ⊙ denotes element-wise multiplication. © represents the concatenation operation.

To address this issue, we propose a selective mutual-guided feature alignment module, which enables selective feature interaction between modalities through a bidirectional cross-attention mechanism and introduces a salient region selection mechanism to focus on the most beneficial feature regions. Specifically, as shown in Fig. 2, given the feature $\mathbf{F}_R \in \mathbb{R}^{C\times H\times W}$ extracted from the visible backbone and the feature $\mathbf{F}_I \in \mathbb{R}^{C\times H\times W}$ extracted from the infrared backbone, we first project them into query ($\mathbf{Q}_i$), key ($\mathbf{K}_i$), and value ($\mathbf{V}_i$), where $i \in \{\mathrm{R}, \mathrm{I}\}$. The alignment process from visible features to infrared features can be formulated as

$$\mathbf{F}_{R\to I} = \mathrm{Softmax}\left(\frac{\mathbf{Q}_\mathrm{R}^\top \otimes \mathbf{K}_\mathrm{I}}{\sqrt{d_k}}\right) \otimes \mathbf{V}_\mathrm{R} + \mathbf{F}_\mathrm{R} \tag{1}$$

where d_k is the scaling factor, ⊗ denotes the matrix multiplication, and Softmax is the Softmax function. Similarly, the alignment process from infrared to visible features is performed symmetrically to obtain $\mathbf{F}_{I\to R}$

$$\mathbf{F}_{I\to R} = \mathrm{Softmax}\left(\frac{\mathbf{Q}_\mathrm{I}^\top \otimes \mathbf{K}_\mathrm{R}}{\sqrt{d_k}}\right) \otimes \mathbf{V}_\mathrm{I} + \mathbf{F}_\mathrm{I}. \tag{2}$$

Subsequently, the aligned features from the two modalities are fused through a concatenation operation and a 1×1 convolution, yielding the base fused feature $\mathbf{F}_{\mathrm{RI}} \in \mathbb{R}^{C\times H\times W}$. After obtaining the base fused features, we design a salient region selection mechanism to suppress redundant information. First, an average pooling (AP) operation is employed to compute the spatial attention map of the fused features. Then, a Top-K selection mechanism is followed to identify and retain the most salient regions. It can be described as

$$\begin{aligned} \mathbf{M}_{\mathrm{avg}} &= \mathrm{AP}\left(\mathbf{F}_{\mathrm{RI}}\right) \\ \mathbf{M}_{\mathrm{filtered}} &= \text{Top-K}\left(\mathbf{M}_{\mathrm{avg}}\right), \quad \mathrm{K} = \lfloor \mathrm{K}_{ratio} \times N \rfloor \end{aligned} \tag{3}$$

where $N = H \times W$ is the total number of spatial locations, K_{ratio} is the retention ratio, which is set to 0.3 in our work, and $\lfloor\cdot\rfloor$ denotes the floor function. The final output features are obtained by an element-wise multiplication (⊙) and residual connection operations, which can be described as

$$\mathbf{F}_{\mathrm{sma}} = \mathbf{F}_{\mathrm{RI}} \odot \mathbf{M}_{\mathrm{filtered}} + \mathbf{F}_{\mathrm{RI}}. \tag{4}$$

SMA effectively mitigates the distribution discrepancy and semantic misalignment between modalities, while employing a salient region selection mechanism to focus on critical information, significantly enhancing the efficiency and robustness of multimodal feature fusion.

C. Enhanced Expert Feature Representation

In RGB-infrared object detection tasks, the characteristics of visible images vary significantly under different lighting conditions and weather patterns, while infrared images exhibit distinct representations due to factors such as thermal target properties and heat distribution patterns. Traditional single-modality feature extraction networks struggle to adapt to these complex modality characteristic variations, resulting in limited feature representation capability.

To address these challenges, we propose an EER module that incorporates a sparse MoEs architecture in both infrared and visible branches, enabling adaptive optimization of dual-modal features through specialized expert networks. As shown in Fig. 3, this module adopts a three-branch input structure, which processes visible features $\mathbf{F}_R \in \mathbb{R}^{C\times H\times W}$ extracted from the visible branch, alignment guidance features $\mathbf{F}_{sma} \in \mathbb{R}^{C\times H\times W}$ generated by the parent SMA module and infrared features $\mathbf{F}_I \in \mathbb{R}^{C\times H\times W}$ extracted from the infrared branch, respectively. First, $\mathbf{F}_R$ and $\mathbf{F}_I$ are fed into two independent sparse MoE modules for deep feature refinement. Unlike dense fully-connected structures that apply all transformations uniformly to every input, sparse MoEs adaptively activate only the most relevant experts for each input. This enables specialized feature processing while avoiding interference and redundancy. Each sparse MoE module contains N expert networks, where each expert consists of two 3×3 convolutional layers, but only the Top-K experts are activated during forward propagation, with $K \le N$, ensuring computational efficiency while maintaining representation capacity. The gating networks for visible and infrared branches dynamically compute expert selection weights through a global AP (GAP) and a Softmax function operation

$$\begin{aligned} \mathbf{w}_R &= \mathrm{Softmax}\left(\mathrm{GAP}\left(\mathbf{F}_R\right)\right) \\ \mathbf{w}_I &= \mathrm{Softmax}\left(\mathrm{GAP}\left(\mathbf{F}_I\right)\right). \end{aligned} \tag{5}$$

For each input feature, only the Top-K experts with the highest weights are activated, where K denotes the sparse

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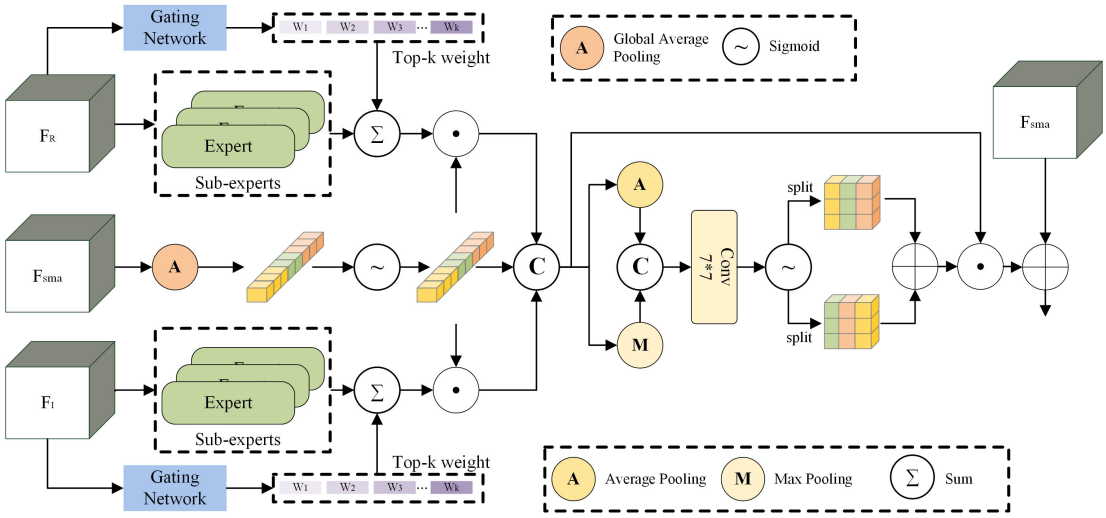


Fig. 3. Architecture of EER.

activation factor. The sparse MoE outputs combine these selected experts

$$\mathrm{MoE}_R(\mathbf{F}_R) = \sum_{i\in\mathcal{T}_R} \mathbf{w}_{Ri} \odot \mathrm{Expert}_i(\mathbf{F}_R)$$
$$\mathrm{MoE}_I(\mathbf{F}_I) = \sum_{i\in\mathcal{T}_I} \mathbf{w}_{Ii} \odot \mathrm{Expert}_i(\mathbf{F}_I) \quad (6)$$

where $\mathcal{T}_R = \text{Top-K}(\mathbf{w}_R)$ and $\mathcal{T}_I = \text{Top-K}(\mathbf{w}_I)$ represent the sets of indices of the selected experts for visible and infrared branches, respectively. Concurrently, the feature $\mathbf{F}_{sma}$ is utilized to generate an alignment guidance attention map

$$\mathbf{A} = \sigma(\mathrm{GAP}(\mathbf{F}_{sma})) \quad (7)$$

where σ represents the Sigmoid function. This attention map $\mathbf{A}$ modulates the MoE-refined visible and infrared features

$$\mathbf{F}'_R = \mathrm{MoE}_R(\mathbf{F}_R) \odot \mathbf{A}, \quad \mathbf{F}'_I = \mathrm{MoE}_I(\mathbf{F}_I) \odot \mathbf{A}. \quad (8)$$

The modulated features are then concatenated and subjected to a dual-pooling aggregation operation, employing both AP and max pooling (MP), to capture complementary spatial contexts

$$\mathbf{F}_{\text{fuse}} = \mathrm{Concat}\left(\mathbf{F}'_R, \mathbf{F}'_I\right), \quad \mathbf{F}_{avg} = \mathrm{AP}(\mathbf{F}_{\text{fuse}})$$
$$\mathbf{F}_{max} = \mathrm{MP}(\mathbf{F}_{\text{fuse}}), \quad \mathbf{F}_{\text{concat}} = \mathrm{Concat}\left(\mathbf{F}_{avg}, \mathbf{F}_{max}\right). \quad (9)$$

A spatial refinement is then performed. Specifically, spatial attention weights are first computed by applying a 7×7 convolution followed by a sigmoid activation to the concatenated features. These weights are then applied through element-wise multiplication to recalibrate the feature maps. Finally, the refined features are fused with the original alignment features via a residual connection, producing the enhanced output

$$\mathbf{S} = \sigma\left(\mathrm{Conv}_{7\times7}(\mathbf{F}_{concat})\right)$$
$$\mathbf{F}_{\text{refined}} = \mathbf{F}_{\text{fuse}} \odot \left(\mathbf{S}_{[:,\mathbf{0}]} + \mathbf{S}_{[:,\mathbf{1}]}\right) + \mathbf{F}_{sma}. \quad (10)$$

In summary, the EER module mitigates feature instability via MoE that adapts to varying inputs. Its spatial attention modulation further enables effective cross-modal interaction. This design achieves optimized feature representation and enhancement, providing a reference solution for multimodal tasks.

D. Cross-Channel Aware Feature Enhancement

Similarity measurement plays a pivotal role in exploring the intrinsic relationships between features, as it effectively distinguishes important features and enhances semantically consistent regions [39]. By precisely quantifying the correlations between features, interference from redundant information can be mitigated and more refined feature enhancement can be achieved.

Based on this, we propose a CAE module to achieve refined feature representation. As illustrated in Fig. 4, the input feature map $\mathbf{X} \in \mathbb{R}^{2C\times H\times W}$ is first equally split along the channel dimension into two complementary components $\mathbf{F}_1, \mathbf{F}_2 \in \mathbb{R}^{C\times H\times W}$.

Subsequently, a dual-path structure is utilized, where global features are perceived in parallel through both AP and MP operations. For each feature branch $\mathbf{F}_j$, where $j \in \{1, 2\}$, we compute channel-wise attention vectors by using GAP and global MP (GMP) operations

$$\mathbf{w}_j^{avg} = \mathrm{GAP}\left(\mathbf{F}_j\right), \quad \mathbf{w}_j^{max} = \mathrm{GMP}\left(\mathbf{F}_j\right). \quad (11)$$

These two vectors: 1) $\mathbf{w}_j^{avg}$ and 2) $\mathbf{w}_j^{max}$, are then concatenated and processed by a multilayer perceptron (MLP). The MLP contains three convolutional layers: 1) a 1×1 expansion layer with a ratio of 2; 2) a standard 3×3 layer; and 3) a 1×1 reduction layer. This operation facilitates interaction between the two information pathways, leading to a more robust attention representation $\mathbf{w}_j$. To explicitly model the inter-dependencies between the two feature branches $\mathbf{F}_1$ and $\mathbf{F}_2$, we compute the cosine similarity between their refined attention representations $\mathbf{w}_1$ and $\mathbf{w}_2$. The similarity score $s \in [-1, 1]$ is calculated as

$$s = \frac{\langle \mathcal{N}(\mathbf{w}_1), \mathcal{N}(\mathbf{w}_2) \rangle}{\|\mathcal{N}(\mathbf{w}_1)\|_2 \cdot \|\mathcal{N}(\mathbf{w}_2)\|_2} \quad (12)$$

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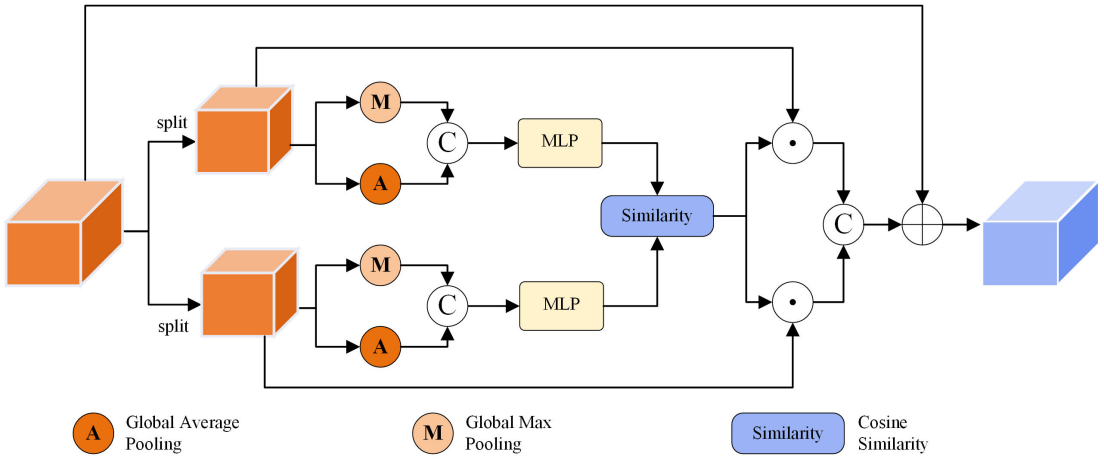


Fig. 4. Architecture of CAE.

where $\mathcal{N}(\cdot)$ denotes the L2 normalization, $\|\cdot\|_2$ computes the Euclidean distance, and $\langle\cdot,\cdot\rangle$ denotes the inner product. This score s serves as a gating mechanism to quantify the semantic consistency within channels. Finally, the $\mathbf{F}_1$ and $\mathbf{F}_2$ are modulated and fused using this similarity score to produce the enhanced output $\mathbf{Y}$. It can be described as

$$\mathbf{F}_1^{out} = \mathbf{F}_1 \odot s, \quad \mathbf{F}_2^{out} = \mathbf{F}_2 \odot s$$
$$\mathbf{Y} = \text{Concat}\left(\mathbf{F}_1^{out}, \mathbf{F}_2^{out}\right) + \mathbf{X}. \tag{13}$$

The proposed CAE module establishes an effective cross-channel interaction mechanism. This architecture enables the model to adaptively amplify semantically consistent regions while suppressing feature conflicts, ultimately leading to more refined and discriminative feature representations for cross-modal tasks.

IV. Experiments

A. Datasets

To validate the efficacy of our approach, we conduct evaluations on two widely used RGB-Infrared object detection benchmarks: 1) the VEDAI [19] and 2) DroneVehicle [8] datasets. These datasets are characterized by their paired multimodal imagery and diverse aerial scenes, providing a comprehensive testbed for assessing cross-modal fusion capabilities under challenging real-world conditions.

1) VEDAI: VEDAI is a multimodal object detection dataset from aerial imagery. It provides high-resolution aerial images with annotations over 3600 instances, covering various vehicle types, such as cars, trucks, and vans. The dataset contains over 1200 matched infrared-visible image pairs, of which 1089 pairs are for training and 121 pairs for testing. All objects are annotated with horizontal bounding boxes (HBBs).

2) DroneVehicle: DroneVehicle is a large-scale benchmark dataset constructed for multimodal object detection from an UAV perspective. It provides 28 439 pairs of optical and infrared images, which are divided according to standard benchmarks into 17 990 training sets, 1469 validation sets, and 8980 test sets. The dataset includes five categories: car, bus, truck, freight car, and van. It encompasses diverse lighting conditions, geographical locations, and viewing angles, aiming to accurately replicate the complexity and diversity of real-world scenarios. Furthermore, all object instances in the dataset are annotated with oriented bounding boxes (OBBs).

B. Implementation

All of our experiments were conducted on a single NVIDIA GeForce RTX 4090 GPU using the PyTorch framework, with Ubuntu 22.04 and CUDA version 12.8. Models were trained for 300 epochs with a batch size of 8, optimized by Adam at an initial learning rate of 0.01, momentum of 0.937, and weight decay of 0.0005. During inference, nonmaximum suppression (NMS) was applied with an IoU threshold of 0.7. For the VEDAI and DroneVehicle datasets, the image size for both training and inference was set to 640×640. All experiments were carried out with consistent hyperparameter configurations.

Model performance was evaluated using standard detection metrics: mean average precision at 50% IoU (mAP_{50}) and the more comprehensive mAP across IoU thresholds from 0.5 to 0.95 ($\text{mAP}_{50:95}$). Additionally, parameter count (Params), floating-point operations (FLOPs), and frames per second (FPS) were reported to reveal model complexity and computational efficiency.

C. Performance Comparison

To systematically evaluate the performance of the proposed model, we conducted comparative experiments with multiple state-of-the-art methods on the VEDAI dataset, including YOLOFusion [40], SuperYOLO [41], MROD-YOLO [42], ACDF-YOLO [43], DEYOLO [44], CFT [45], ICAFusion [28], and YOLO-FIV [27]. Simultaneously, comprehensive comparisons were made with current advanced approaches on the DroneVehicle dataset, such as CRSIOD [46], YOLOFIV [27], M2FNet [47], MDPMFN [25], ADMPF [26], MGMF [48], and GAANet [29].

1) Experiments on the VEDAI Dataset: We comprehensively evaluated the proposed SCFDet on the VEDAI dataset

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TABLE I
EXPERIMENTS ON THE VEDAI DATASET. * INDICATES THAT THE EXPERIMENTAL RESULTS ARE REPRODUCIBLE

Method	modality	Car	Pickup	Camping	Truck	Other	Tractor	Boat	Van	mAP_{50}	$mAP_{50:95}$
YOLOFusion	I	86.70	75.90	66.60	77.10	43.00	62.30	70.70	84.30	74.00	-
	R	91.10	82.30	75.10	78.30	33.30	81.20	71.80	62.20	74.30	-
	R+I	91.70	85.90	78.90	78.10	54.70	71.90	71.70	75.20	78.60	49.14
SuperYOLO	I	87.90	81.39	76.90	61.56	39.39	60.56	46.08	71.00	65.60	
	R	90.30	82.66	76.69	68.55	53.86	79.48	58.08	70.30	72.49	-
	R+I	91.13	85.66	79.30	70.18	57.33	80.41	60.24	76.50	75.09	-
MROD-YOLO	I	89.60	80.20	75.30	75.00	44.60	68.90	59.40	78.50	73.00	47.30
	R	93.20	81.60	70.30	69.50	61.00	72.60	66.50	70.00	75.80	46.90
	R+I	**94.00**	**89.70**	75.00	74.80	73.20	72.10	68.00	76.20	77.90	49.80
ACDF-YOLO	I	85.73	80.51	73.82	70.07	32.65	58.66	48.71	83.96	66.76	41.48
	R	88.32	84.90	76.80	80.74	61.20	80.74	52.43	65.65	71.91	41.56
	R+I	89.84	87.92	78.52	75.31	67.33	76.27	73.66	75.97	78.10	47.88
DEYOLO*	R+I	88.02	79.65	62.04	81.03	52.68	77.54	68.12	**90.63**	74.96	48.32
CFT*	R+I	87.41	73.31	59.55	75.29	65.54	59.53	78.19	69.44	71.03	42..49
ICAFusion*	R+I	89.16	80.35	48.94	**83.37**	71.02	78.48	**81.63**	65.12	74.76	44.16
YOLOFIV	R+I	93.89	87.42	**82.10**	80.13	60.84	**82.15**	75.47	79.82	80.16	-
SCFDet(Ours)	R+I	90.83	83.73	75.60	75.51	**75.88**	80.47	73.99	87.62	**80.45**	**50.60**

and compared it with state-of-the-art multimodal detectors. As detailed in Table I, the performance declines sharply when models use only the RGB (R) or infrared (I) dataset, while the R+I fusion brings clear and stable improvement. This performance difference stems primarily from the inherent susceptibility of single-modal data to environmental interference. RGB images exhibit limited discriminability under poor illumination, while infrared images often lack clear textural definition. In contrast, multimodal fusion effectively harnesses their complementary properties, yielding more robust feature representations that are essential for reliable remote sensing detection.

Through further analysis of the detection performance across different categories, we found that SCFDet performs exceptionally well in the other category, achieving an AP_{50} score of 75.88%, which reflects the method's robustness in handling infrequent objects without fixed patterns. Meanwhile, SCFDet also achieves the second-best performance in the tractor, boat, and van categories. It is noteworthy that although the AP_{50} value of SCFDet for certain individual categories (e.g., car, pickup) is slightly lower than the other SOTA methods, it achieves the optimal balance in the overall mAP_{50} metric, reflecting the method's balanced detection performance across all categories. Overall, our proposed SCFDet achieved the best performance of 80.45% on mAP_{50}, surpassing all compared methods. Particularly on the more challenging $mAP_{50:95}$ metric, SCFDet achieved a score of 50.60%, significantly outperforming the other methods, demonstrating its robust detection capability across various IoU thresholds.

To visually compare the detection performance, Fig. 5 shows the qualitative results from the baseline model, DEYOLO, and our proposed SCFDet on the VEDAI dataset. In scenes with relatively uncluttered and homogeneous backgrounds, both the baseline and DEYOLO achieve a reasonable level of detection accuracy with few erroneous predictions. In contrast, under the challenging conditions of urban settings characterized by intricate structural backgrounds and significant object appearance variations, the baseline model exhibits a noticeable number of false detections. This suggests its limited robustness when confronted with complex scenes. Meanwhile, our SCFDet consistently delivers reliable performance across all scenarios, demonstrating the dynamic feature extraction capability of the EER module, which adaptively enhances feature representations under varying conditions.

In summary, SCFDet demonstrates exceptional comprehensive detection performance on the VEDAI dataset, particularly showcasing robustness when handling complex scenarios and diverse target categories, which validates the advancement and practicality of the proposed method for multimodal remote sensing object detection tasks.

2) Experiments on the DoneVehicle Dataset: All methods evaluated on the DroneVehicle dataset are multimodal, leveraging both RGB and infrared data. As shown in Table II, the results demonstrate that the proposed method, SCFDet, achieves state-of-the-art performance. It attains the highest overall mAP_{50} of 83.60%, outperforming the other advanced methods such as ADMPF (82.05%) and MDPMFN (81.26%). Furthermore, SCFDet excels in three specific categories, achieving the best performance in bus (97.87%), truck (85.26%), and van (71.02%). It also records a significantly higher $mAP_{50:95}$ of 68.69%, which is notably better than MGMF (55.20%) and GAANet (52.10%), indicating its superior robustness across various detection criteria. The superior performance of SCFDet can be attributed to its synergistic module design. The SMA module effectively aligns cross-modal features. The EER module dynamically adapts feature transformations according to varying illumination and thermal contrast conditions. Meanwhile, the CAE module enhances semantic consistency across channels, thereby improving discrimination.

While SCFDet shows exceptional performance in most categories, it is noteworthy that GAANet achieves the best result for detecting freight cars, and MDPMFN performs best on car detection. This highlights the varying strengths

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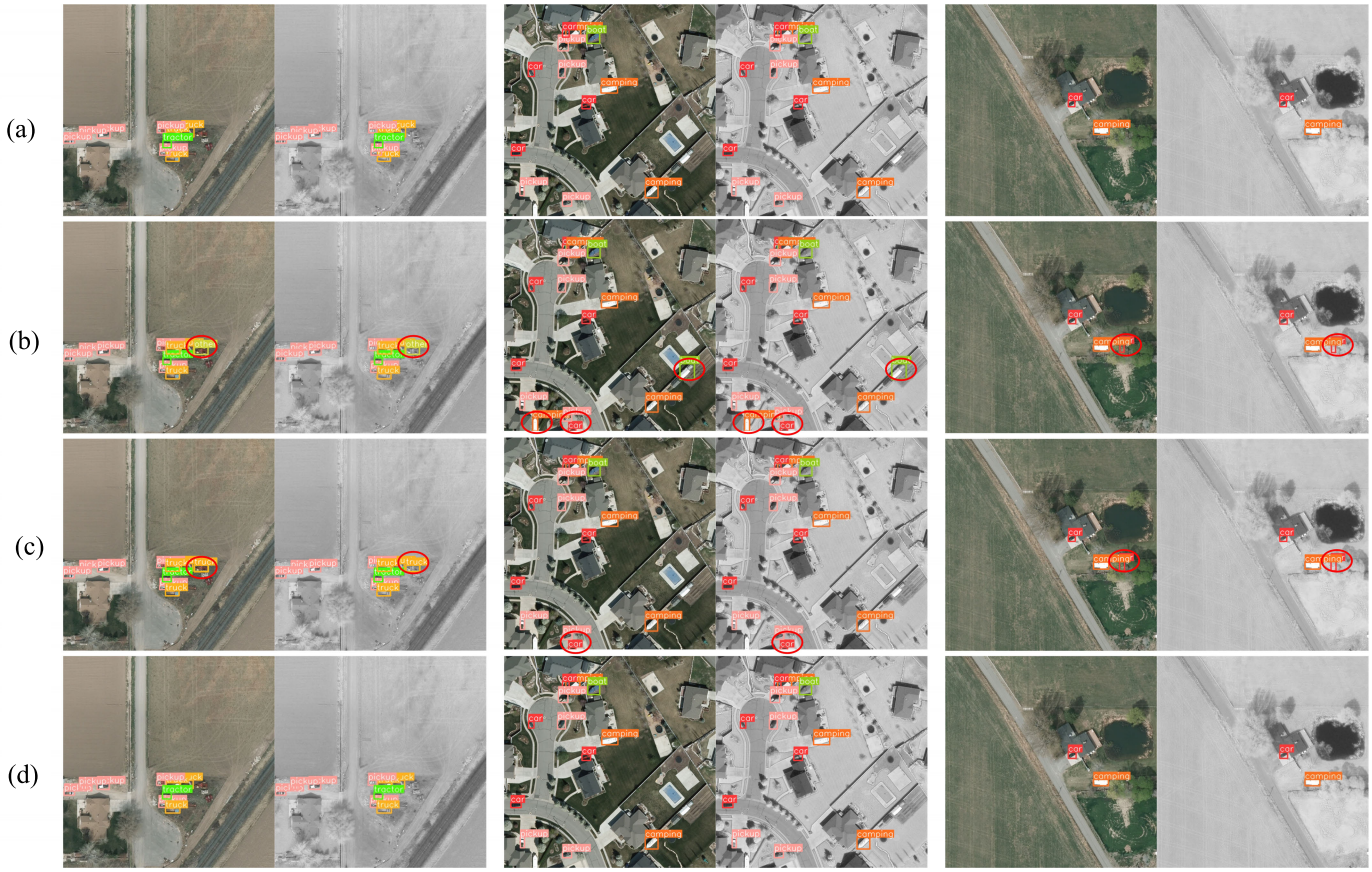


Fig. 5. Detection results on the VEDAI dataset. The top row presents ground-truth bounding boxes for reference. Prediction results from baseline and DEYOLO are displayed in the second and third rows, respectively. The bottom row illustrates detections generated by our proposed SCFDet. The red circle denotes the false detection. (a) GT. (b) Baseline. (c) DEYOLO. (d) SCFDet.

TABLE II
EXPERIMENTS ON THE DRONEVEHICLE DATASET

Method	modality	Car	Bus	Truck	Freight car	Van	mAP_{50}	$mAP_{50:95}$
CRSIOD	R+I	94.58	92.02	63.67	46.29	45.74	68.46	-
YOLOFIV	R+I	95.89	91.56	64.23	34.57	64.71	64.71	-
M2FNet	R+I	-	-	-	-	-	71.50	-
MDPMFN	R+I	**97.95**	95.50	80.87	67.68	64.33	81.26	-
ADMPF	R+I	97.94	95.93	82.28	69.34	64.77	82.05	-
MGMF	R+I	91.40	91.10	70.10	78.50	69.40	80.30	55.20
GAANet	R+I	90.33	80.51	68.86	**89.82**	65.55	79.01	52.10
SCFDet(Ours)	R+I	97.87	**97.87**	**85.26**	66.78	**71.02**	**83.60**	**68.69**

of different models across specific object types. Overall, the results confirm the effectiveness of the proposed SCFDet framework for multimodal object detection in drone-captured scenarios.

Furthermore, we present qualitative results of SCFDet on the DroneVehicle dataset in Fig. 6. In complex road environments with dense and diverse object categories, the model maintains high classification accuracy, which primarily benefits from the CAE module. By enhancing semantically consistent features and suppressing conflicting responses across channels, the CAE module refines the feature representation, leading to more discriminative capability for category distinction. More critically, under degraded illumination, such as fog or complete darkness, the model reliably overcomes visual limitations and delivers stable detection performance. This robustness highlights effective collaboration between SMA and EER. SMA ensures focused alignment of salient regions, while EER adaptively enhances features via dynamic expert selection. Collectively, they enable reliable fusion even under severe imaging constraints.

In summary, SCFDet achieves state-of-the-art performance on the DroneVehicle dataset, demonstrating superior detection accuracy across various object categories and environmental conditions. The proposed model exhibits strong robustness in complex scenes and under challenging illumination, validating its effectiveness for multimodal object detection in aerial scenarios.

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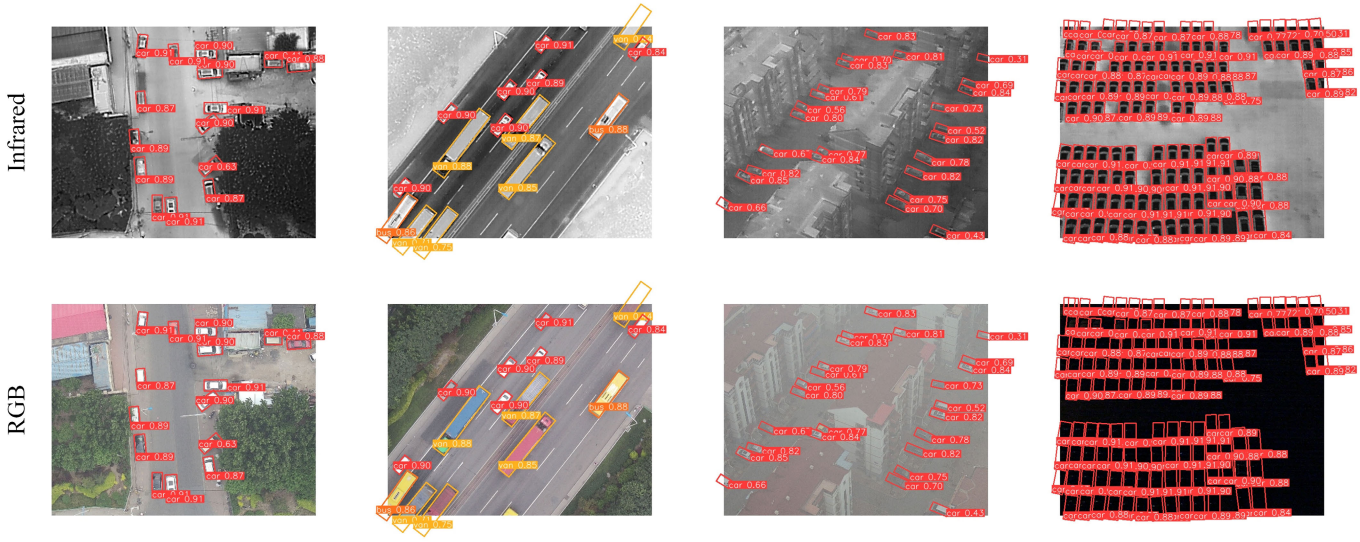


Fig. 6. Results of SCFDet on the dronevehicle dataset, shown on the infrared and visible images, respectively.

TABLE III
COMPLEXITY AND EFFICIENCY COMPARISON OF DIFFERENT METHODS

Method	Type	FLOPs(G)	Params(M)	FPS
MROD-YOLO	HBB	227.3	45.35	–
DEYOLO	HBB	–	48.78	45.14
CFT	HBB	–	44.53	36.90
ICAFusion	HBB	–	120.28	30.44
YOLOFIV	HBB	**54.96**	**10.49**	–
SCFDet (Ours)	HBB	130.12	34.76	42.42
ADMPF	OBB	**68.80**	**31.80**	–
GAANet	OBB	194.90	79.00	–
SCFDet (Ours)	OBB	133.81	35.61	34.25

TABLE IV
RESULTS OF ABLATION EXPERIMENT ON THE VEDAI DATASET

Baseline	SMA	EER	CAE	mAP_{50}	$mAP_{50:95}$
✓				75.73	47.23
✓	✓			77.00	48.70
✓	✓	✓		79.74	49.68
✓	✓	✓	✓	80.45	50.60

D. Complexity Analysis

To comprehensively evaluate the performance of SCFDet, this section compares it with other state-of-the-art methods in terms of Params, FLOPs, and FPS. As shown in Table III, in the HBB detection task, SCFDet demonstrates a computational complexity of 130.12 GFLOPs and 34.76 M parameters, with an inference speed of 42.42 FPS. Compared to similar methods, YOLOFIV achieves the highest computational and parameter efficiency (54.96 GFLOPs/10.49 M Params), while SCFDet maintains a moderately competitive level. It demonstrates significantly lower computational load than MROD-YOLO (227.3 GFLOPs) and requires substantially fewer parameters than ICAFusion (120.28 M), which reflects reasonable complexity control in its design. In terms of FPS, SCFDet outperforms CFT and ICAFusion, and is comparable to DEYOLO.

In the more challenging OBB detection task, SCFDet maintains a similar complexity profile (133.81 GFLOPs/35.61 M Params/34.25 FPS). Although its computational demand is higher than the ADMPF (68.80 GFLOPs), it remains notably lower than GAANet (194.90 GFLOPs). Moreover, SCFDet's parameter count is only about 45% of GAANet's (79.00 M), demonstrating superior parameter efficiency.

Comprehensive analysis indicates that SCFDet achieves a balanced trade-off between model performance and computational efficiency across both HBB and OBB tasks, with moderate computational complexity. Particularly noteworthy is its unified architecture, which maintains stable complexity when handling different detection tasks.

E. Ablation Study

A systematic ablation study was conducted on the VEDAI dataset to validate the contribution of each proposed module. The results, summarized in Table IV, demonstrate a progressive performance improvement, confirming the efficacy of our design. The baseline model achieves an mAP_{50} score of 75.73% and an $mAP_{50:95}$ score of 47.23%. After introducing the SMA module, which achieves precise cross-modal feature alignment through bidirectional cross-attention and enhances fusion quality by focusing on critical spatial areas, the model's performance improves significantly, reaching mAP_{50} of 77.00% and $mAP_{50:95}$ of 48.70%.

A more substantial performance boost is observed upon integrating the EER module, which improves the mAP_{50} score to 79.74% and the $mAP_{50:95}$ score to 49.68%. This significant leap is attributed to the module's dynamic adaptation to varying input characteristics. By employing a sparse MoEs architecture, the EER module selectively activates specialized expert networks for optimal feature transformation. Coupled

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TABLE V
COMPARISON OF SMA PERFORMANCE UNDER DIFFERENT K_{ratio} SETTINGS ON THE VEDAI DATASET

K_ratio	mAP_{50}	$mAP_{50:95}$
0.20	75.23	47.26
0.25	77.59	47.55
0.30	**80.45**	**50.60**
0.35	77.62	48.77
0.40	77.56	48.69

TABLE VI
COMPARISON OF EER PERFORMANCE UNDER DIFFERENT N AND K SETTINGS ON THE VEDAI DATASET

K	N	mAP_{50}	$mAP_{50:95}$
1	2	75.95	47.94
	3	80.45	50.60
	4	76.33	48.53
2	2	76.27	48.03
	3	72.55	44.58
	4	75.15	46.84

with spatial attention modulation guided by the preceding SMA output, it significantly enriches the feature representation and enhances robustness against modal instability.

The complete SCFDet model, which further incorporates the CAE module, achieves the peak performance of 80.45% in mAP_{50} and 50.60% in $mAP_{50:95}$. The incorporation of CAE enables a more refined feature fusion by explicitly modeling inter-channel relationships. Through consistency evaluation, it adaptively amplifies semantically coherent features while suppressing feature conflicts across channels. The cumulative improvements observed throughout the ablation stages demonstrate that the SMA, EER, and CAE modules work in synergy, each addressing a distinct facet of the cross-modal fusion challenge to collectively enable robust and accurate object detection.

F. Analysis of SMA

We systematically evaluated the model performance under different K_{ratio} values, as shown in Table V. The experiment results reveal that the model performance does not change linearly with K_{ratio} but rather exhibits an optimal range. When K_{ratio} is set to 0.30, the model achieves the best overall performance, with mAP_{50} reaching 80.45% and $mAP_{50:95}$ reaching 50.60%.

As K_{ratio} increases from 0.20 to 0.30, both metrics show significant improvement, with mAP_{50} increasing by 5.22% and $mAP_{50:95}$ by 3.34%. This trend suggests that at lower ratios, the model may lose some critical features due to excessive filtering, limiting its performance. However, when K_{ratio} exceeds 0.30 and increases further to 0.35 and 0.40, performance noticeably declines, with mAP_{50} dropping to 77.62% and 77.56%, respectively. This indicates that an excessively high retention ratio introduces too much redundant information or noise, interfering with the model's judgment and thus reducing detection accuracy.

In summary, the K_{ratio} parameter in the SMA module requires fine-tuning. This experiment identifies 0.30 as the optimal value for the given task, ensuring that the attention mechanism most effectively focuses on the most discriminative feature regions. This finding provides an important parameter configuration reference for subsequent research.

G. Analysis of EER

To investigate the impact of the sparse MoE configuration within the EER module, we evaluate the model performance under varying numbers of total experts (N) and activated experts (K). The experimental results, summarized in Table VI, reveal a complex and nonlinear relationship between model capacity and detection accuracy, underscoring that simply increasing resources is insufficient for optimal performance.

The configuration achieves its peak performance when $N = 3$ and $K = 1$, yielding the highest scores of 80.45% in mAP_{50} and 50.60% in $mAP_{50:95}$. This optimal setting suggests that for our task, maintaining a moderate number of experts while activating only the most relevant one for each input is the most effective strategy. It allows the model to benefit from specialized feature transformations without introducing excessive computational complexity or potential interference from less relevant experts. In contrast, increasing the total number of experts to $N = 4$ while keeping $K = 1$ leads to a noticeable performance drop, with mAP_{50} decreasing to 76.33%. This decline implies that an overly large expert pool might lead to optimization difficulties if the model capacity exceeds the requirements of the task.

Furthermore, the performance generally degrades when activating more experts ($K = 2$) across all settings of N. The most significant performance collapse occurs for the configuration $N = 3, K = 2$, where mAP_{50} and $mAP_{50:95}$ plummet to 72.55% and 44.58%, respectively. This trend indicates that simultaneously combining multiple transformation paths can introduce feature conflicts or redundancy, which disrupts the formation of a clean and discriminative representation. The results strongly advocate for a sparse activation strategy ($K = 1$) within the EER module, as it effectively harnesses the benefits of expert specialization while avoiding the noise associated with aggregating multiple expert outputs.

V. DISCUSSION

All experiments in our work confirm that the proposed SCFDet framework serves as an effective solution for RGB-infrared object detection in remote sensing. By introducing a selective cross-modal fusion strategy, SCFDet achieves robust and accurate detection performance across diverse aerial scenarios. The core innovation lies in three carefully designed modules. The SMA module enables precise spatial and semantic alignment between RGB and infrared features through bidirectional cross-attention, while its salient region selection mechanism suppresses less informative areas. The EER module incorporates dynamic routing based on a sparse MoE architecture, allowing the model to adaptively strengthen feature representations under varying environmental conditions. Finally, the CAE module refines the fused features

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by evaluating inter-channel consistency, thereby amplifying semantically coherent information and suppressing feature conflicts.

Despite its strong performance, the proposed method has certain limitations. First, the current model architecture is not lightweight enough for deployment on platforms with stringent computational constraints. Second, the proposed method is currently designed and validated only for bimodal (RGB and Infrared) fusion. It remains unexplored whether the proposed selective fusion paradigm can be effectively scaled to scenarios involving more modalities, such as those incorporating LiDAR, SAR, or hyperspectral data. Third, while the Top-K mechanism used in SMA improves computational efficiency by selecting only the most important spatial regions, it may introduce information loss risks, particularly in small object detection scenarios. Specifically, selecting only the Top-K most salient features for fusion could potentially discard subtle features that are critical for representing small objects. These features might not rank highly but are nonetheless essential for detecting small or partially occluded targets.

Future work will focus on several directions. We plan to develop lightweight variants of SCFDet to enhance its suitability for real-time applications on edge devices. Furthermore, we will extend the selective fusion paradigm to support more than two modalities, which requires designing scalable cross-modal interaction mechanisms and dynamic gating policies to handle increased modal diversity. Third, to mitigate the potential information loss caused by the Top-K mechanism, we will explore adaptive thresholding strategies or soft selection mechanisms that preserve low-salience features for small object regions. In addition, applying the proposed fusion philosophy to other multimodal remote sensing tasks, such as segmentation and change detection, represents a promising avenue for future exploration.

VI. Conclusion

In this article, we propose SCFDet, a novel framework for RGB-infrared object detection in remote sensing that leverages a selective cross-modal fusion strategy to address the challenges of feature misalignment and insufficient representation in existing methods. By integrating bidirectional feature alignment, dynamic representation enhancement, and inter-channel feature refinement, SCFDet achieves robust and discriminative feature learning for cross-modal data. Extensive experiments on the VEDAI and DroneVehicle datasets demonstrate that SCFDet achieves state-of-the-art performance, exhibiting strong robustness across diverse scenes and challenging conditions. The proposed framework not only advances RGB-Infrared object detection performance but also provides a general and effective paradigm for feature-level multimodal fusion. Future work will focus on developing lightweight variants and extending the selective fusion strategy to other multimodal remote sensing tasks.

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